How can we utilize those numbers to perform grounding? In very simple terms, we can analyze the log probabilities associated with each token and determine how confident the model is in its answers, which helps us decide whether to trust those answers or seek additional information.

We first establish thresholds for these log probabilities to determine when a response is reliable enough to be considered "grounded." For example, if the log probability for a particular response is above a certain threshold (e.g., −1.0), we might decide that it's trustworthy. On the other hand, if the model indicates low confidence in its answer (e.g., log probabilities suggest uncertainty), we can either

- Prompt the model to provide an alternative answer.
- Use external data sources to supplement or verify the information.
- Ask the user for more context to improve response accuracy.

The following listing shows how to use log probability grounding in the existing RAG application we've developed so far.

Listing 7.11 Integrating log probability grounding

```
// import { RAG } from '@/lib/RAG';
import { LogProbsRag as RAG }
from '@/lib/LogPropsRag';

export async function continueConversation(input) {
  const aiState = getMutableAIState();

  try {
    // Ensure RAG system is initialized
    const rag = await initializeRAG();

    // Perform enhanced RAG query
    const result = await rag.performEnhancedRAG(
      input,
      {
        confidenceThreshold: 85,
        fallbackStrategy: 'ask_user'
      }
    );

    // Log confidence metrics if available
    if (result.confidenceMetrics) {
      console.log('Confidence Metrics:', result.confidenceMetrics);
      console.log('Results:', result);
    }
    // rest of code
  } catch (error) {
    console.error('Error in continuing conversation:', error);
  }
}
```

Comments this out as it shows the original import statement for the RAG class

Uncomments this class to use the updated RAG with log probabilities

Calls the performEnhancedRAG method on the RAG instance, which executes a query with enhanced capabilities

Sets a confidence threshold of 85% for determining if the response is reliable

Specifies a fallback strategy to ask the user for more context if confidence is low